

Cloud Computing Case Study – Online Learning Platform

Section A: Understanding the Need for Cloud

Q1. Explain why cloud computing is suitable for the given startup

Ans: Cloud computing is highly suitable for an online learning platform startup due to the following reasons:

- **Scalability:**
Online learning platforms experience unpredictable user traffic. During exam seasons, thousands of students may access live classes, recorded lectures, and quizzes simultaneously. Cloud computing allows automatic scaling of resources such as servers and storage to meet this increased demand. When traffic reduces during holidays or off-peak periods, the cloud can scale down resources, ensuring efficient utilization.
- **Cost Efficiency:**
Startups generally operate with limited budgets. Traditional IT infrastructure requires heavy upfront investment in hardware, data centers, and maintenance. Cloud computing eliminates these capital expenses by providing a pay-as-you-go model, where the startup only pays for the resources it actually uses.
- **Speed to Market:**
Cloud platforms provide ready-to-use services such as video streaming, databases, analytics, and content delivery networks. This enables the startup to launch its platform quickly without spending time and resources on building infrastructure from scratch.

Section B: Cloud Deployment Model Analysis

Q2. Choosing the Cloud Type

a) Which cloud type would you recommend for this startup?

- Public Cloud
- Private Cloud
- Hybrid Cloud

Ans: Public Cloud

b) Justify your choice based on:

- **Cost:**
Public cloud services do not require capital expenditure. The startup pays only for the resources consumed, making it affordable for early-stage businesses.
- **Scalability:**
Public cloud providers offer virtually unlimited computing and storage resources, allowing the platform to scale instantly based on user demand.

- **Security:**
Leading public cloud providers such as AWS, Microsoft Azure, and Google Cloud implement strong security measures including encryption, identity management, and compliance certifications, which are sufficient for a new startup.
- **Operational Overhead:**
The cloud provider manages physical servers, networking, and hardware maintenance. This allows the startup to focus on application development and user experience rather than infrastructure management.

Q3. Business Growth Scenario

After two years, the startup:

- Handles sensitive student data
- Partners with universities
- Requires higher compliance and control

a) Would you continue with the same cloud type?

Ans: No, I would likely transition to a Hybrid Cloud.

b) If not, which cloud type would you suggest and why?

Ans: A **Hybrid Cloud** is the most suitable option at this stage.

- A hybrid cloud combines both public and private cloud environments.
- Sensitive student records, examination data, and university information can be stored in a private cloud to ensure higher security, compliance, and control.
- At the same time, public cloud services can still be used for hosting learning content, video streaming, and handling large user traffic efficiently.

Section C: Cloud Service Model Evaluation

Q4. Choosing the Service Model

a) Which cloud service model is most suitable for developing this platform?

- IaaS
- PaaS
- SaaS

Ans: The most suitable cloud service model for developing this platform is **PaaS (Platform as a Service)**.

b) Explain why the other service models are less suitable for this use case.

Ans:

- **IaaS (Infrastructure as a Service):**
1. IaaS requires the startup to manage operating systems, middleware, networking, and security configurations.

2. This increases technical complexity and operational workload, which may distract the team from core application development.

- **SaaS (Software as a Service):**

1. SaaS provides ready-made applications but does not allow sufficient customization.
2. The startup needs flexibility to build unique features such as custom quizzes, learning analytics, and interactive dashboards, which SaaS cannot provide.
3. PaaS provides a balanced solution by managing infrastructure and runtime environments while allowing developers to focus on building and deploying applications.

Q5. Shared Responsibility Model

For the selected service model, specify who is responsible (Cloud Provider or User) for managing the following:

Components	Responsible Party
• Infrastructure	Cloud Provider
• Operating System	Cloud Provider
• Application code	User(Startup)
• Scaling	Cloud Provider(Automated)
• Security patches	Cloud Provider(For OS/Infra)

Section D: Scalability & Performance

Q6. Handling Fluctuating Traffic

Explain how cloud computing helps the platform handle:

Cloud computing provides several features to handle varying traffic loads effectively:

- **Sudden Spikes in User Traffic:**
During exam periods, cloud platforms use **auto-scaling** to automatically add server instances when user demand increases. This ensures uninterrupted access to the platform.
- **Reduced Traffic During Off-Peak Periods:**
During low usage periods, **elasticity** allows the cloud system to release unused resources, minimizing costs.

Relevant Cloud Features:

- **Auto-scaling:** Automatically adjusts resources based on demand.
- **Elasticity:** Enables seamless expansion and contraction of resources.
- **Load Balancing:** Distributes incoming traffic across multiple servers to prevent system overload.

Section E: Cost Optimization

Q7. Cost Management in Cloud

a) Explain how the pay-as-you-go pricing model benefits this startup

Ans: Pay-as-you-go Benefits:

The pay-as-you-go pricing model offers significant benefits to the startup:

- The startup pays only for the resources it uses, avoiding expenses on idle infrastructure.
- It reduces financial risk during early stages of business growth.
- Saved capital can be reinvested into improving learning features, content quality, and user experience.

Conclusion

Cloud computing provides a flexible, scalable, and cost-effective foundation for an online learning platform startup. By initially using a public cloud with a PaaS model and later transitioning to a hybrid cloud, the startup can support rapid growth, ensure data security, and manage costs efficiently. Cloud features such as auto-scaling, elasticity, and pay-as-you-go pricing make cloud computing an ideal solution for modern EdTech platforms.